EWB CLOUD LABS

AWS VPC Peering Project Report

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Batch: AWS Batch 1

Region: ap-south-1 (Mumbai)

Completion date: 24 October 2026

Project objective

Configure private communication between two Amazon VPCs using VPC Peering, validate forward and reverse connectivity, demonstrate a controlled security-group failure and recovery, and access a private EC2 instance through a bastion host.

1. Architecture and resource inventory

The lab uses two VPCs in the Mumbai region. VPC A contains a public bastion subnet and a private application subnet. VPC B contains a private application subnet and has no Internet Gateway attached. VPC peering provides private routing between the two VPCs.

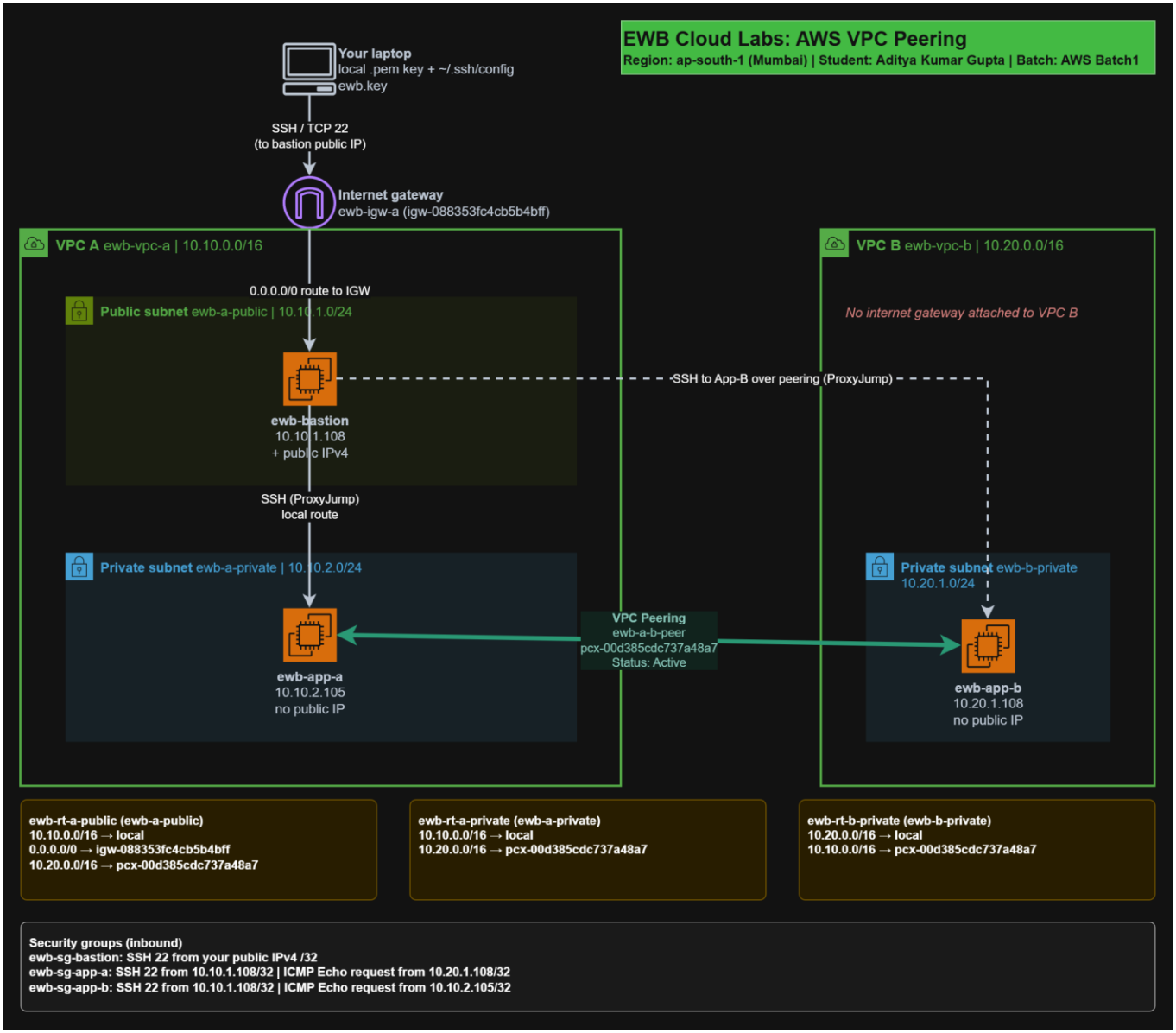


Figure 1. Submitted architecture diagram. Some route-table peering IDs are placeholders where not specified.

Component	Name / CIDR / address
Region	ap-south-1 (Mumbai)
VPC A	ewb-vpc-a — 10.10.0.0/16
VPC A public subnet	ewb-a-public — 10.10.1.0/24
VPC A private subnet	ewb-a-private — 10.10.2.0/24
VPC B	ewb-vpc-b — 10.20.0.0/16
VPC B private subnet	ewb-b-private — 10.20.1.0/24
Internet Gateway	ewb-igw-a (attached to VPC A)
VPC peering	ewb-a-b-peer — shown Active in supplied screenshot
Bastion EC2	ewb-bastion — 10.10.1.108; public IPv4 used for SSH
App A EC2	ewb-app-a — 10.10.2.105; no public IPv4
App B EC2	ewb-app-b — 10.20.1.108; no public IPv4
Key pair	ewb-lab

2. Routing and security configuration

The route tables use local routes for each VPC CIDR and peering routes for the remote VPC CIDR. The public route table also has a default route to the Internet Gateway. The supplied final route-table screenshots are evidence of the configured routes.

Route table	Subnet association	Required routes
ewb-rt-a-public	ewb-a-public	10.10.0.0/16 → local; 0.0.0.0/0 → IGW; 10.20.0.0/16 → peering
ewb-rt-a-private	ewb-a-private	10.10.0.0/16 → local; 10.20.0.0/16 → peering
ewb-rt-b-private	ewb-b-private	10.20.0.0/16 → local; 10.10.0.0/16 → peering

Inbound security-group rules shown in the architecture

- ewb-sg-bastion: SSH (TCP 22) from the user's public IPv4 address with a /32 source.
- ewb-sg-app-a: SSH (TCP 22) from bastion private IP 10.10.1.108/32; ICMP Echo Request from App B 10.20.1.108/32.
- ewb-sg-app-b: SSH (TCP 22) from bastion private IP 10.10.1.108/32; ICMP Echo Request from App A 10.10.2.105/32.

Security principle: SSH access to private instances is restricted to the bastion's private IP. ICMP is limited to the peer application's private IP for connectivity testing.

3. Test plan and observed results

The results below correspond to the supplied screenshots and the user's confirmation of the test outcomes.

Test	Observed result	Evidence file
Baseline ping before peering/routes	Failed; baseline capture shows 100% packet loss.	18-baseline-ping-fail.png
VPC peering connection	Pending state captured, followed by Active state.	19-peering-pending.png; 20-peering-active.png
Forward ping: App A → App B	Succeeded; screenshot supplied.	24-ping-forward(AppA-to-AppB).png
Reverse ping: App B → App A	Succeeded; screenshot supplied.	25-ping-reverse(AppB-to-AppA).png
SSH from bastion to App B via ProxyJump	Successful App B shell captured.	26-ssh-app-b.png
Controlled failure	App B inbound ICMP rule removed; ping from A failed.	27-failure-rule-removed(ICMP from SG).png; 28-failure-ping-fails (From A to B).png
Recovery	ICMP rule restored; ping succeeded again.	29-rule-restored.png; 30-recovery-ping-ok.png

Interpretation

Successful forward and reverse pings demonstrate that both VPCs had routes to the other VPC and that the relevant security-group ICMP rules permitted the tests. The controlled failure isolated the App B inbound ICMP rule: removing it caused the ping to fail, and restoring it allowed the ping to succeed. SSH to App B through the bastion demonstrates the ProxyJump access path without assigning App B a public IPv4 address.

Troubleshooting lesson: VPC peering does not automatically configure routes or security groups. Both VPCs need correct return routes, and security groups must permit the required traffic.

4. Cleanup and cost awareness

The supplied cleanup evidence includes terminated lab instances, a deleted peering connection, and deleted VPCs. The user confirmed that the remaining lab networking resources were deleted and that AWS Billing was checked after cleanup.

Cleanup item	Evidence / status
EC2 instances	Terminated; screenshot supplied.
VPC peering connection	Deleted; screenshot supplied.
VPCs	Deleted; screenshot supplied.
Subnets, route tables, Internet Gateway and security groups	User confirmed deletion.
EBS volumes, snapshots, Elastic IPs and key pair	User confirmed checks/deletion where applicable.
Billing	User reports Billing was checked after cleanup; no zero-charge claim is made.

Note: AWS billing data can take time to update. Checking Billing does not itself prove that no charges occurred; review relevant service charges for the lab period if final cost confirmation is needed.

5. Learning summary

- Created and organized VPCs and subnets with separate public and private network roles.
- Configured an Internet Gateway and route tables for bastion Internet access and private communication across VPC peering.
- Applied least-privilege inbound security-group rules for SSH and ICMP testing.
- Verified connectivity in both directions and demonstrated how a security-group rule can cause and resolve a connection failure.
- Used a bastion host with SSH ProxyJump to reach a private EC2 instance without a public IP.
- Cleaned up lab resources and checked AWS Billing after completion.

Conclusion

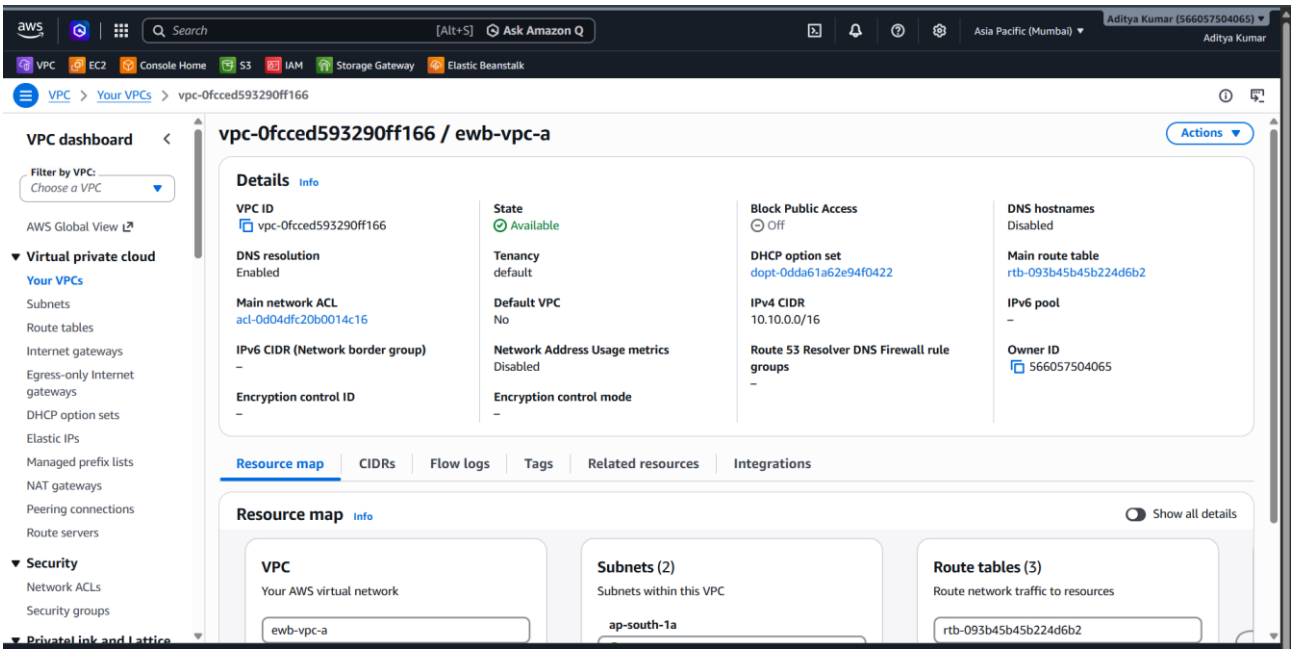
The supplied evidence supports completion of the core VPC peering lab workflow: network configuration, active peering, bidirectional private connectivity, controlled failure and recovery, bastion-based SSH access, and cleanup. This report documents the supplied evidence; it is not an AWS-issued certification or a guarantee that the course provider awards a certificate.

Appendix A — Screenshot evidence

The following pages preserve the screenshots supplied with the report. Each image is labelled with its original filename and purpose.

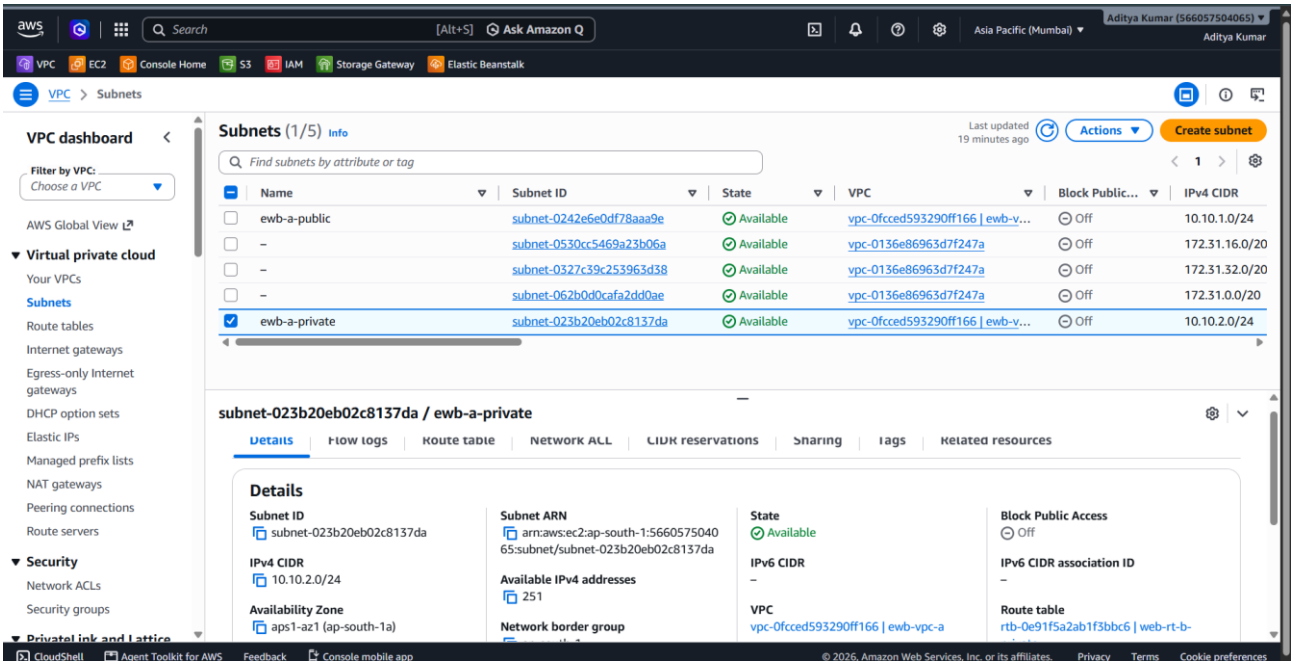
VPC A details

01-vpc-a.png



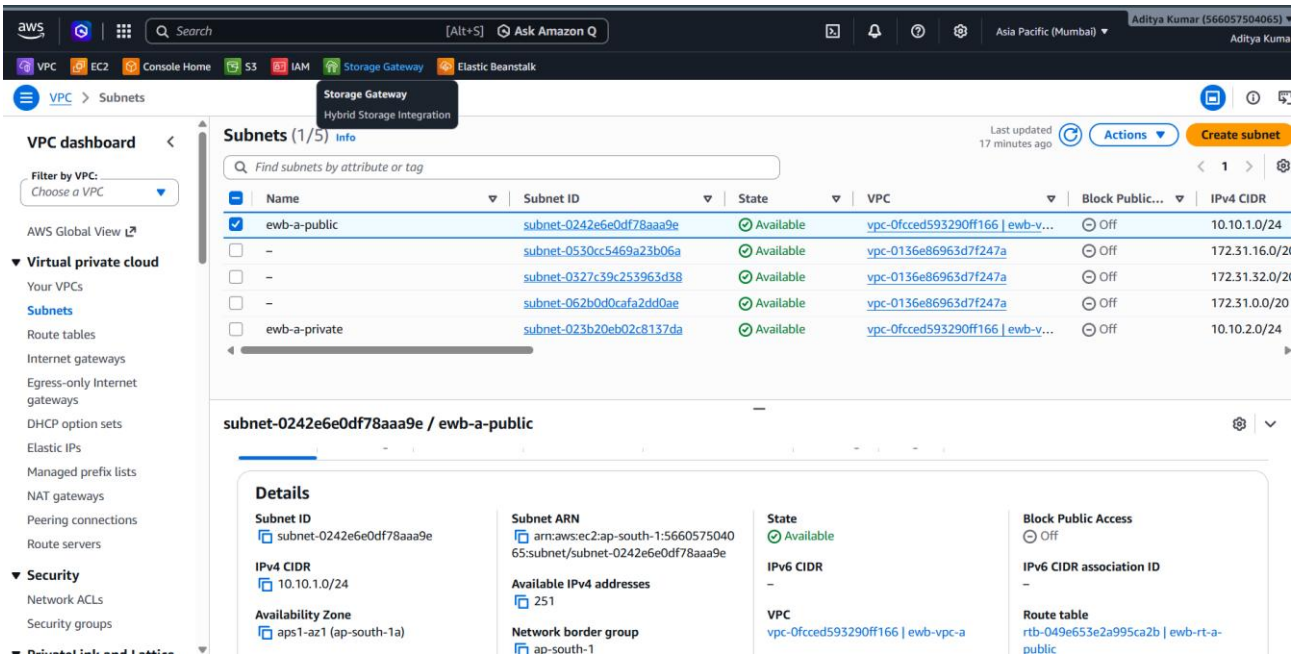
Private subnet in VPC A

02-subnets-a-private.png



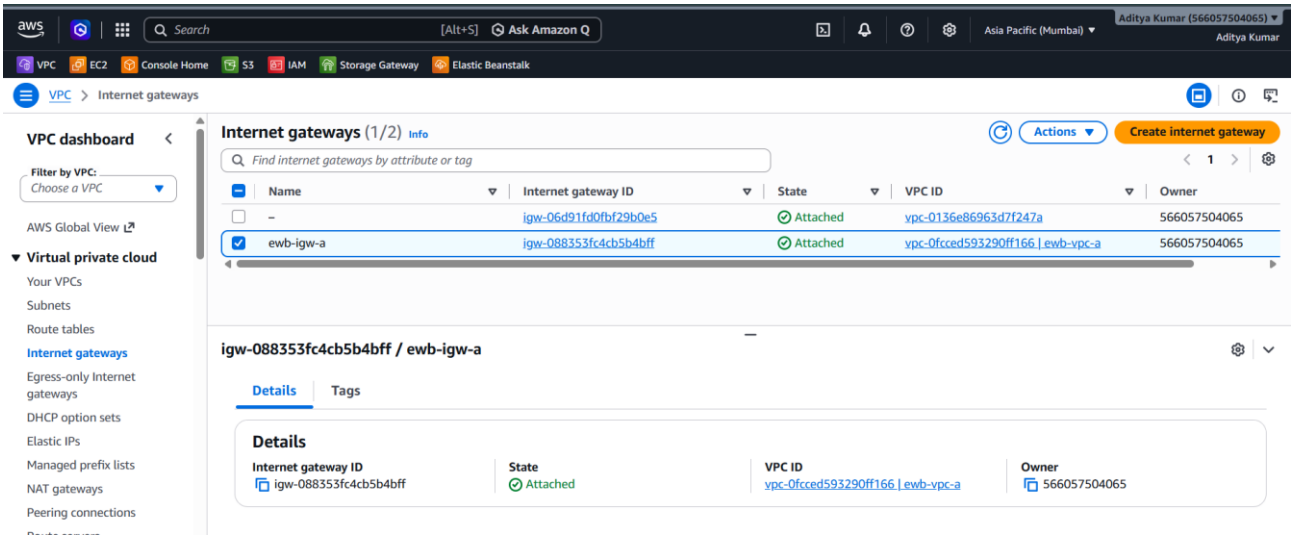
Public subnet in VPC A

02-subnets-a-public.png



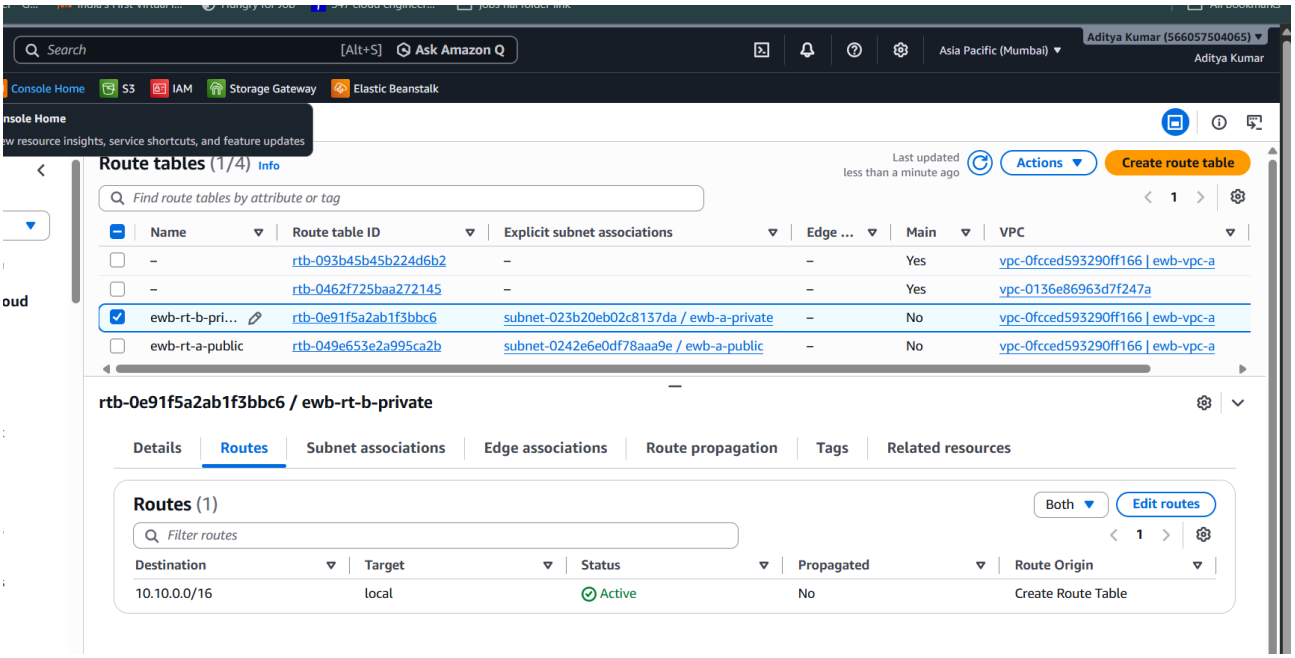
Internet Gateway attached to VPC A

03-igw-attached.png



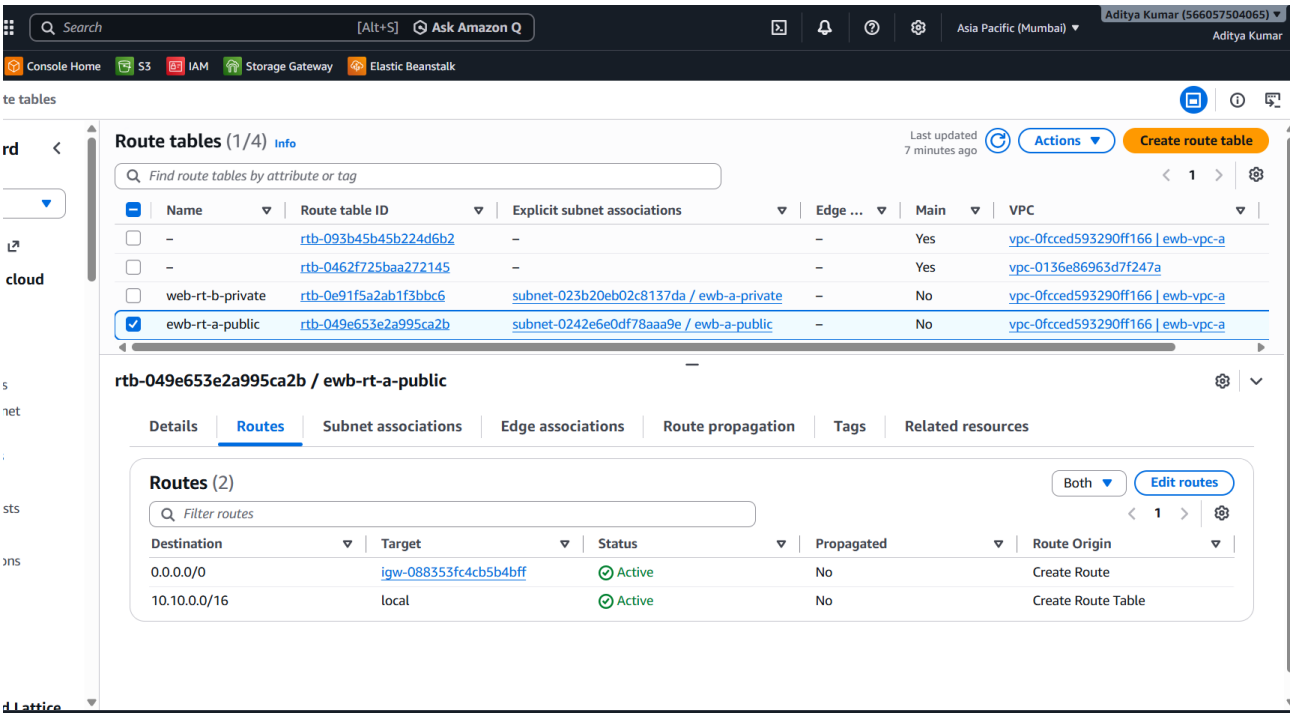
VPC A private route table

04-rt-a-private.png



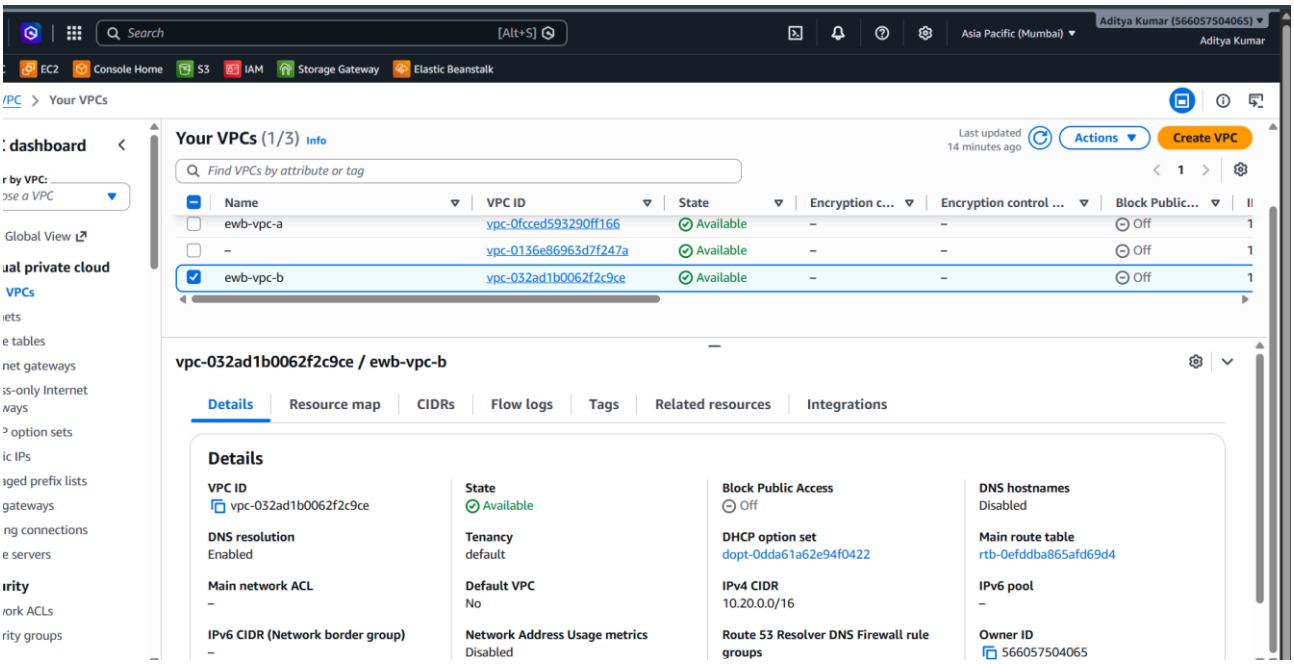
VPC A Public route Table

04-rt-a-public.png



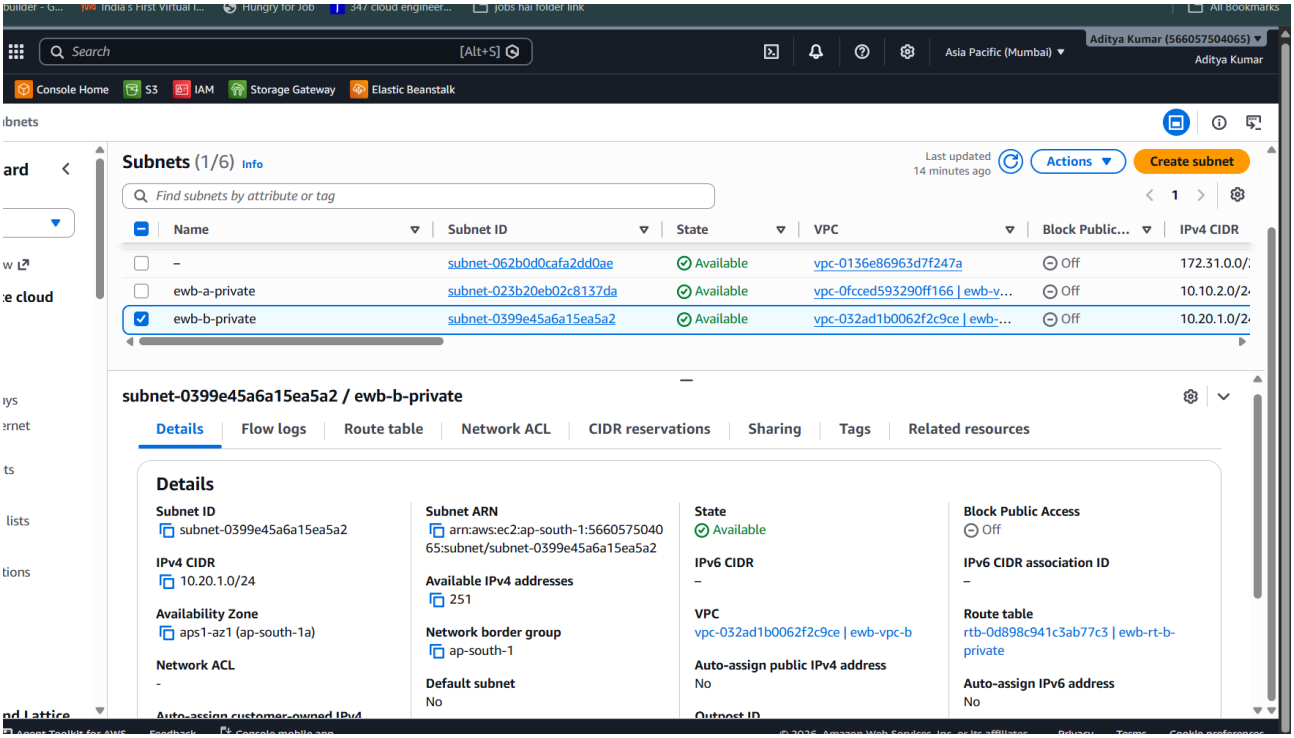
VPC B details

06-vpc-b.png



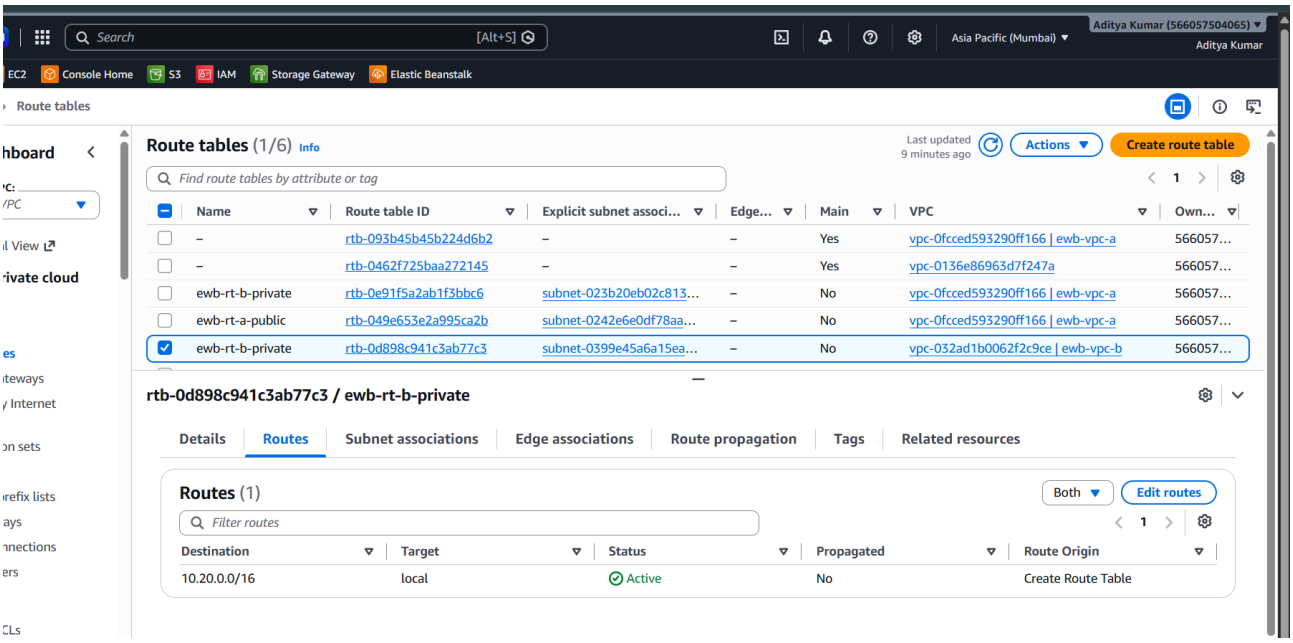
Private subnet in VPC B

07-subnet-b.png



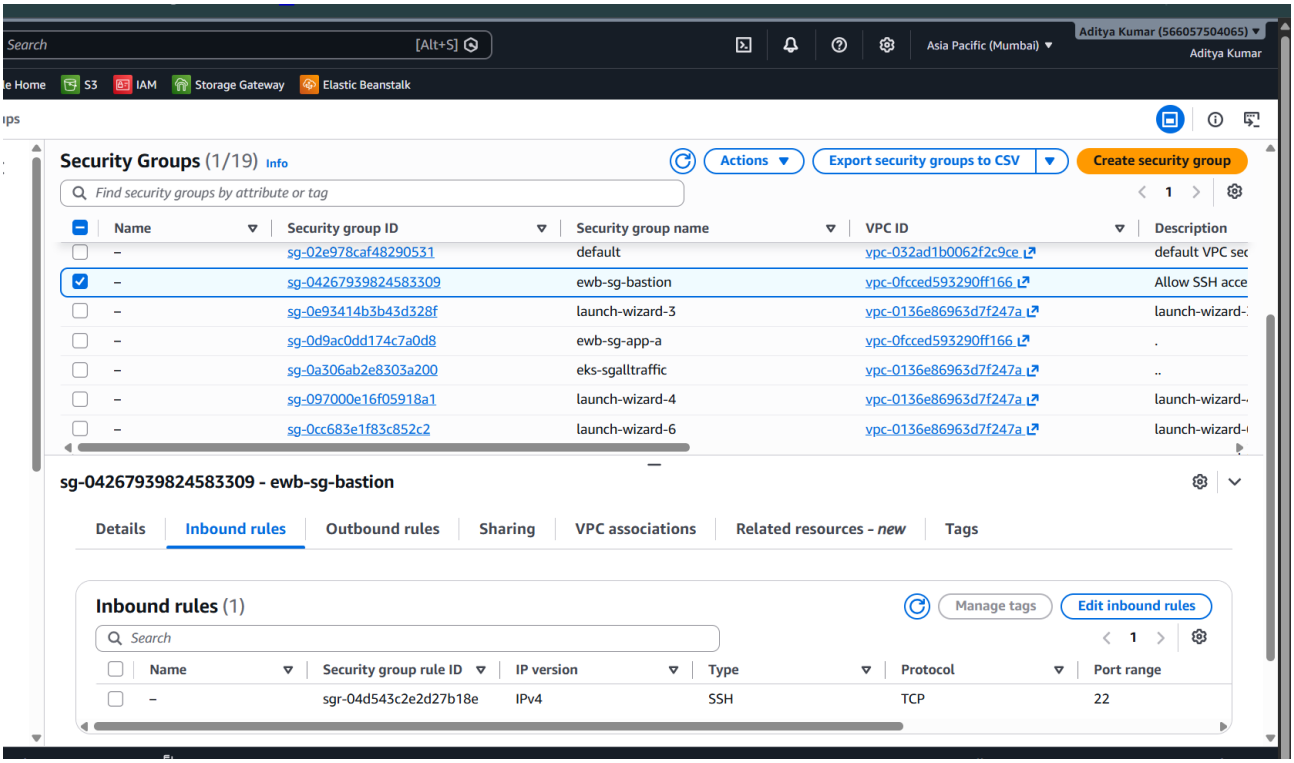
VPC B route table before peering

08-rt-b-before-peering.png



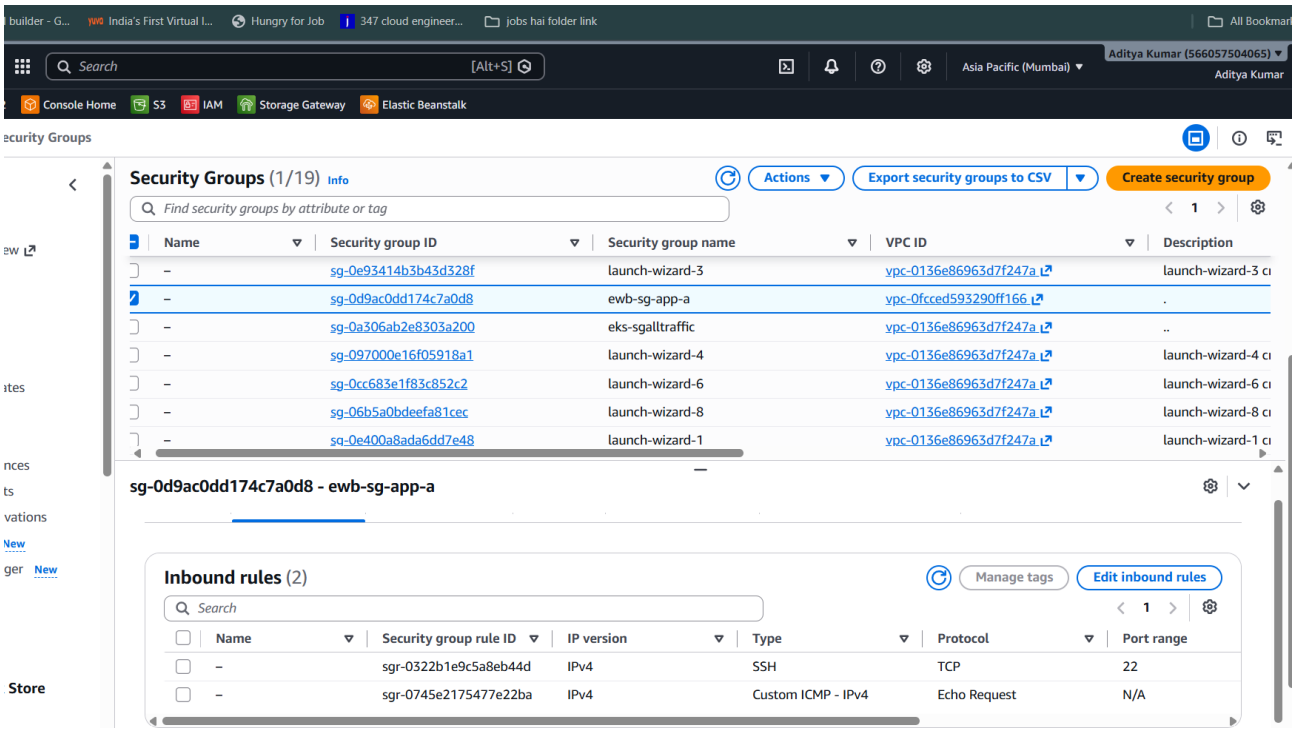
Bastion security-group details(Inbound Details)

09-sg-bastion.png



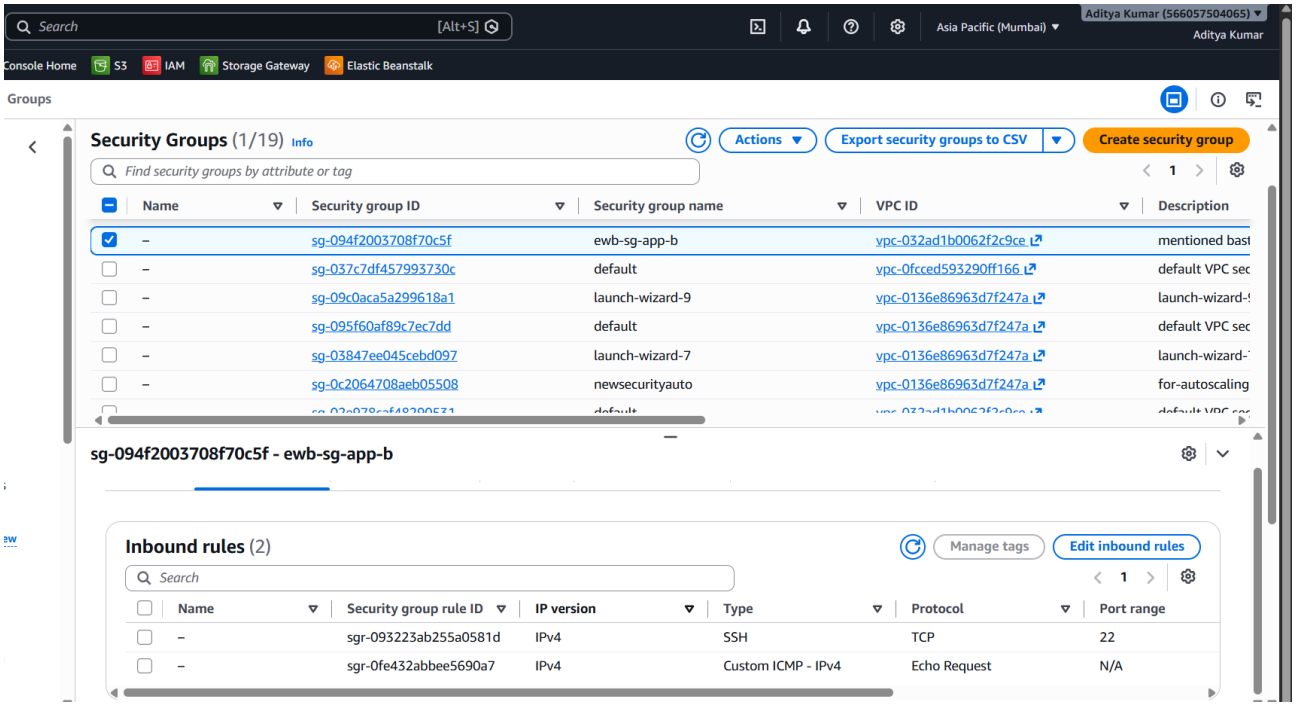
App-A security group(Inbound Rules)

10-sg-app-a.png



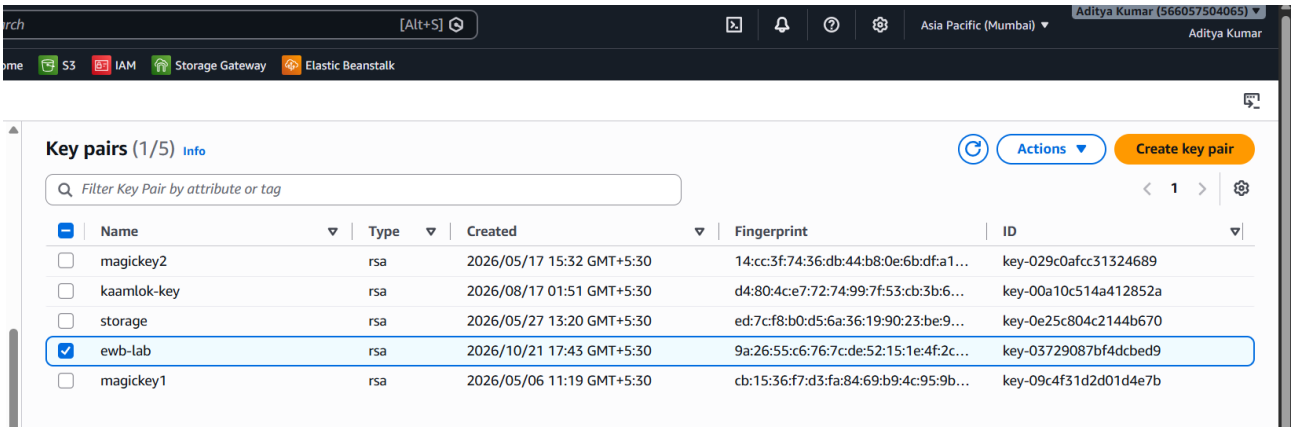
App B security group(Inbound Rules)

11-sg-app-b.png



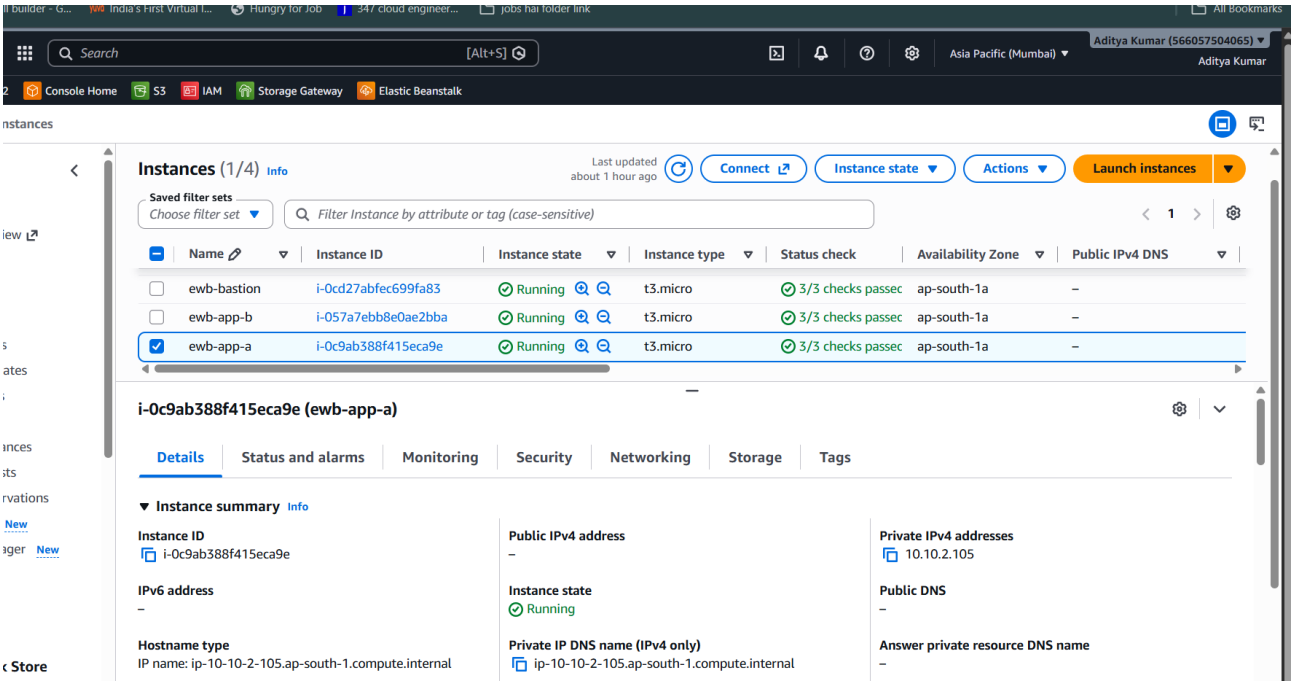
Lab SSH key pair(ewb-lab)

12-keypair.png



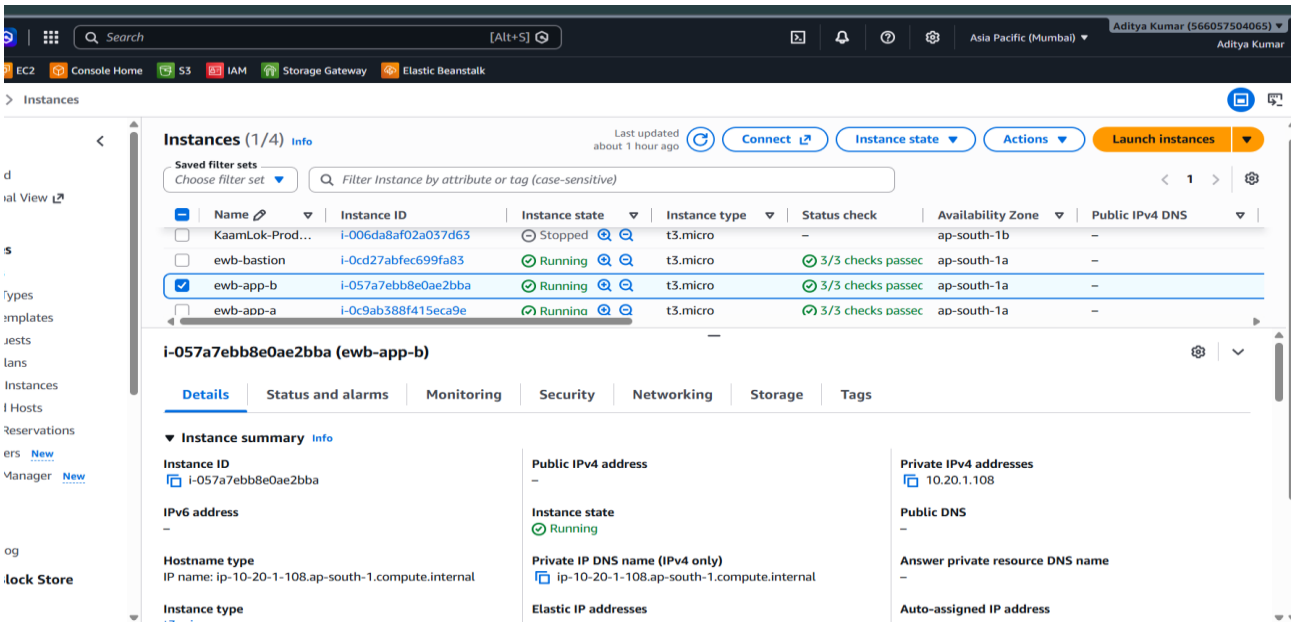
App A EC2 details

13-app-a-details.png



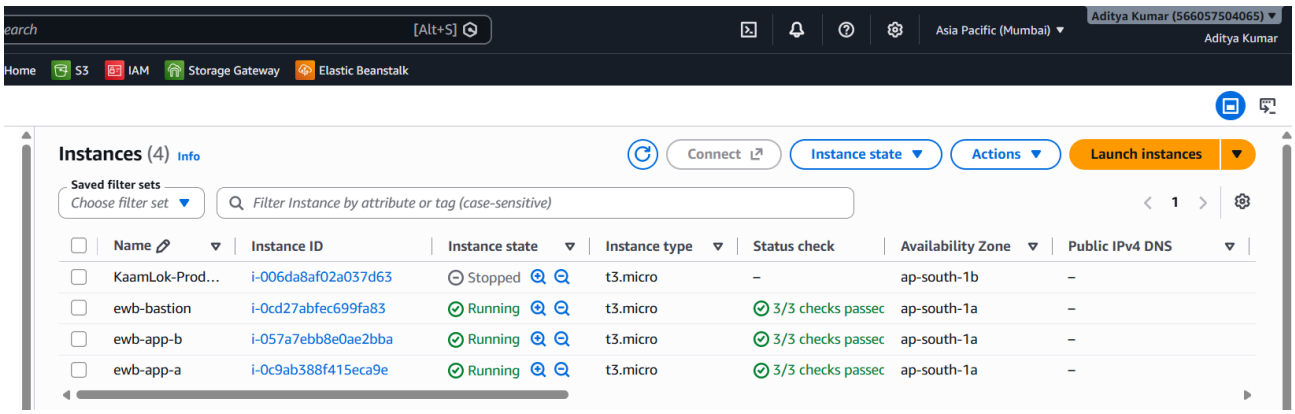
App B EC2 details

14-app-b-details.png



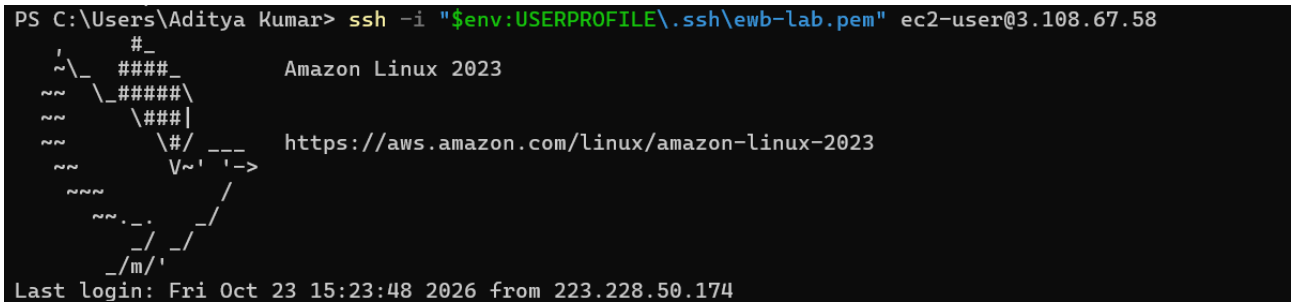
EC2 instances list before cleanup

15-instances-list.png



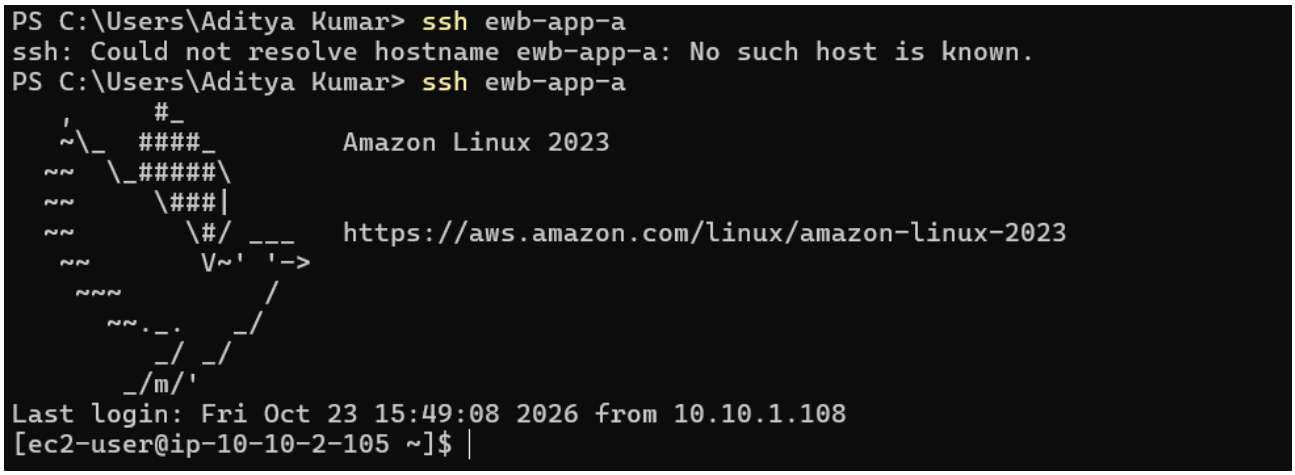
SSH session to bastion

16-ssh-bastion.png



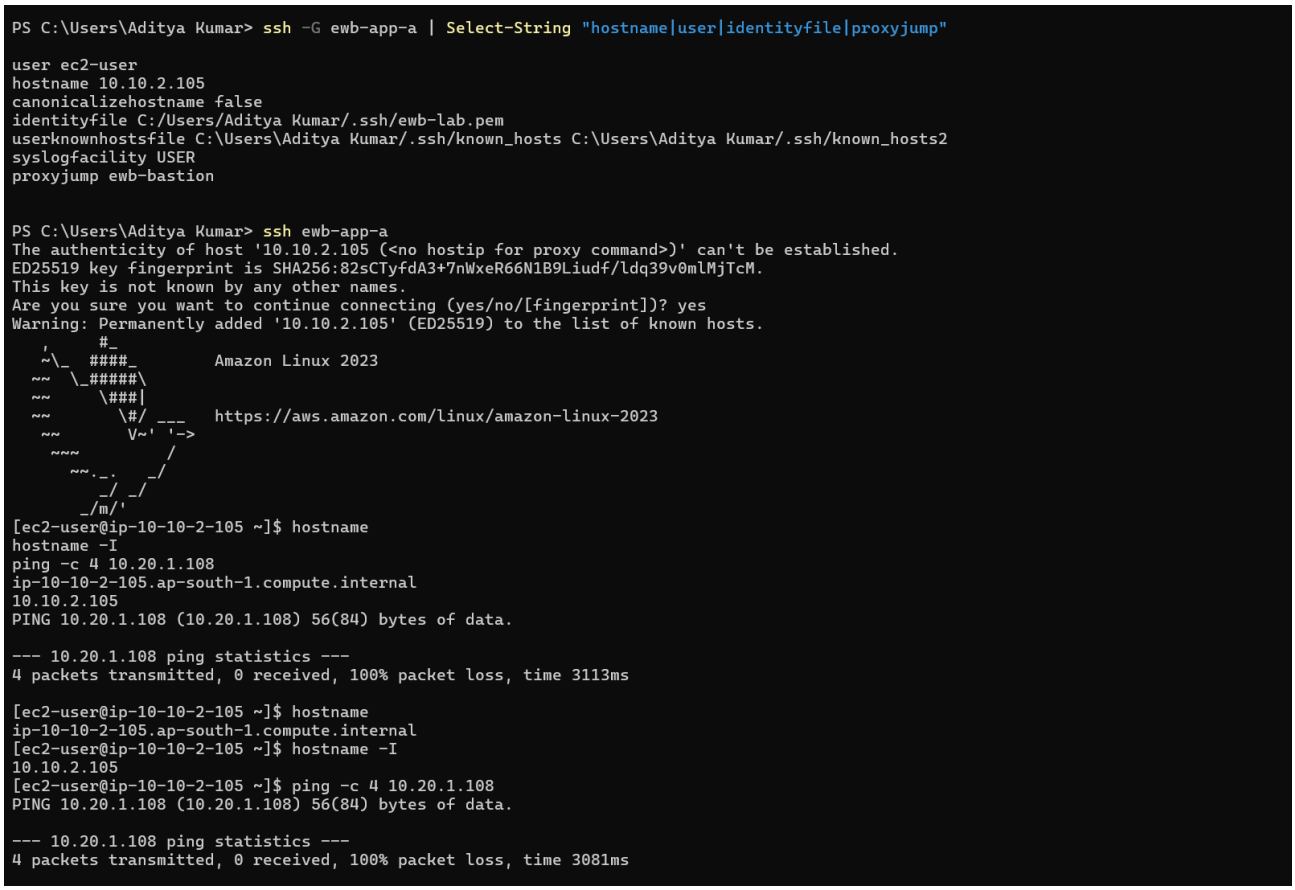
SSH session to App A

17-ssh-app-a.png



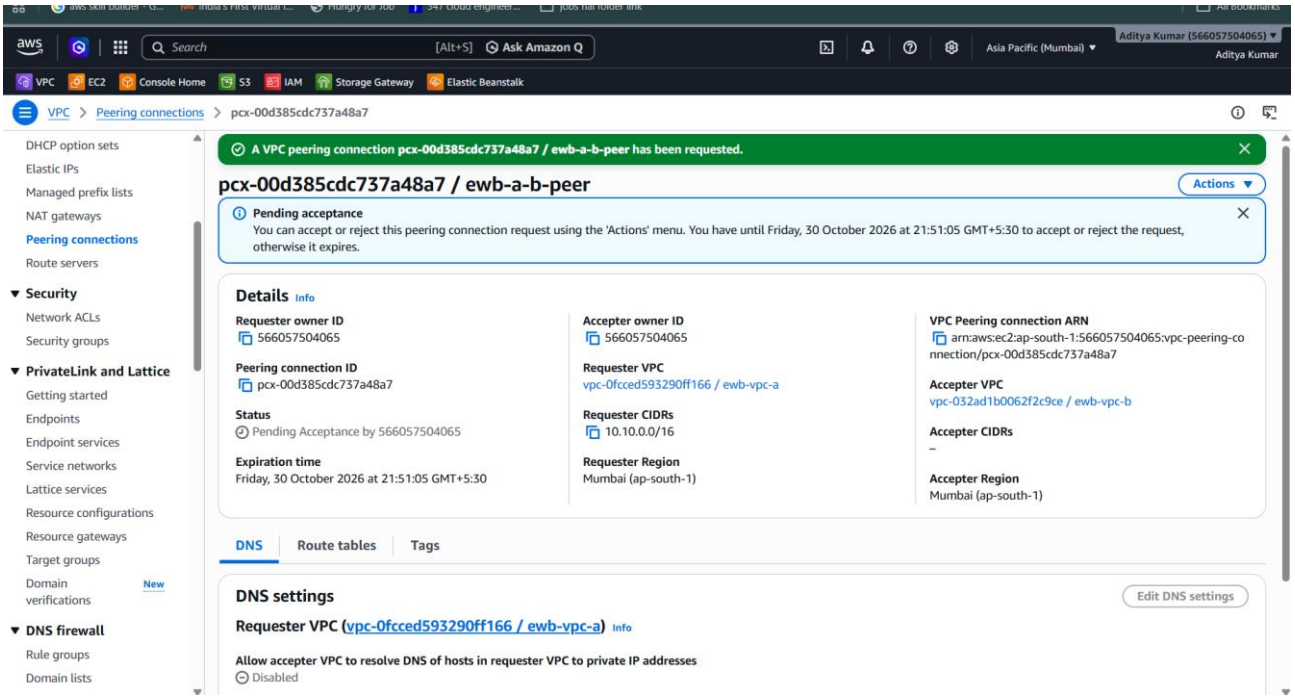
Baseline ping failed before peering/routing was ready

18-baseline-ping-fail.png



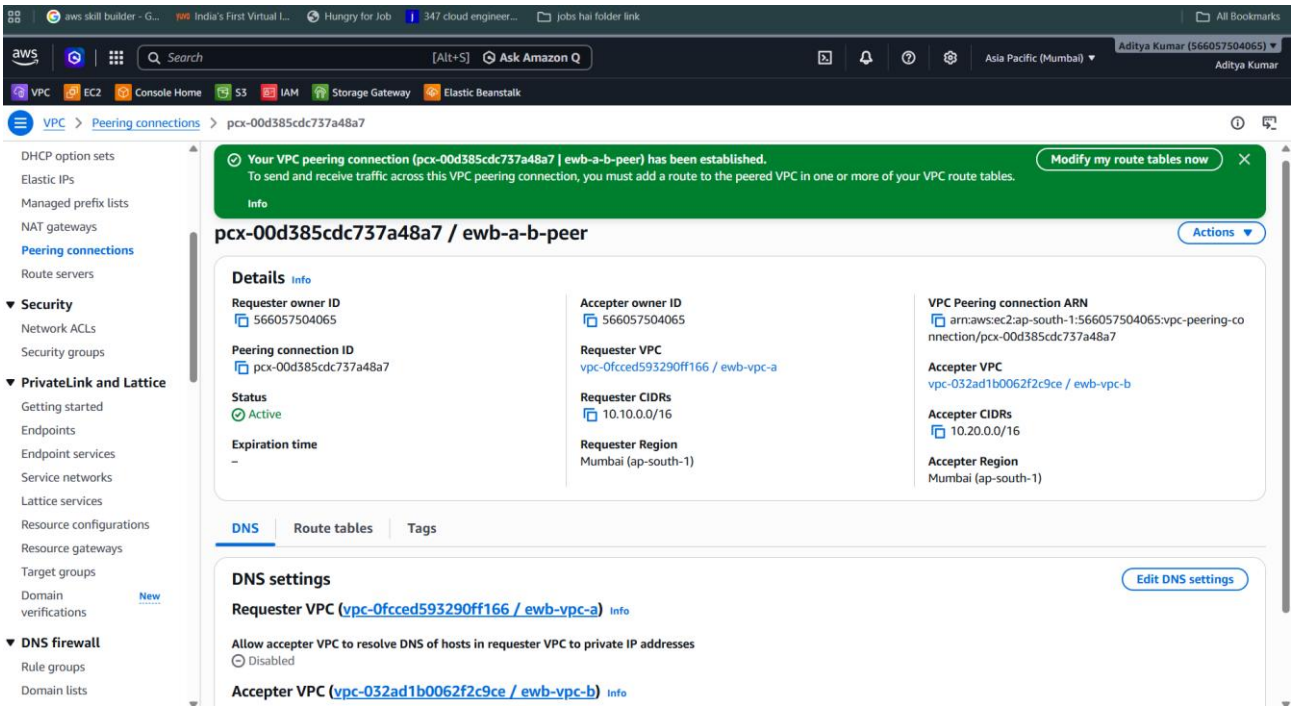
Peering connection pending

19-peering-pending.png



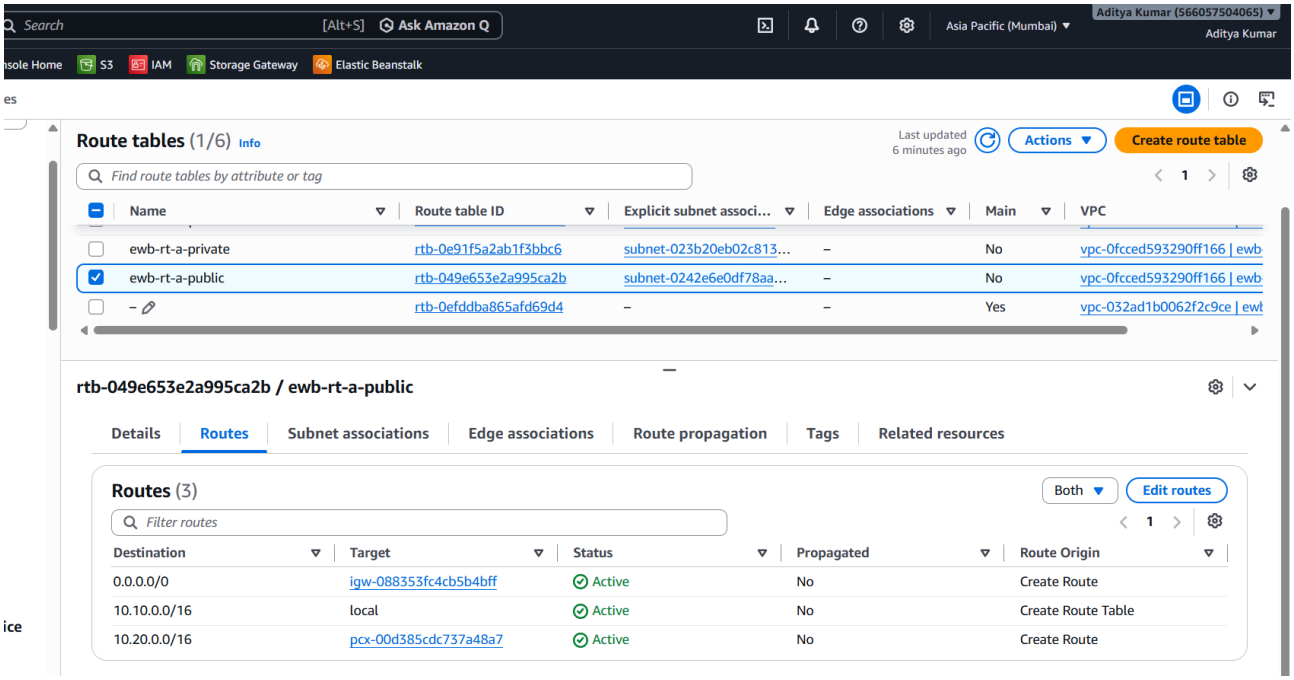
Peering connection active

20-peering-active.png



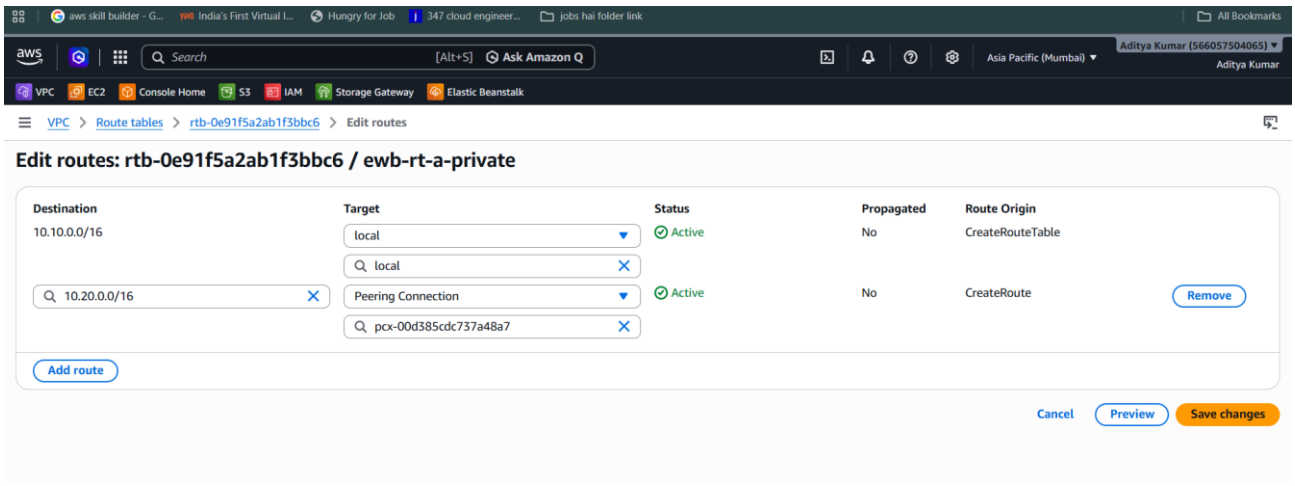
Final VPC A public route table (Adding of Peering connection route in Target with previously present igw and local route)

21-rt-a-public-final.png



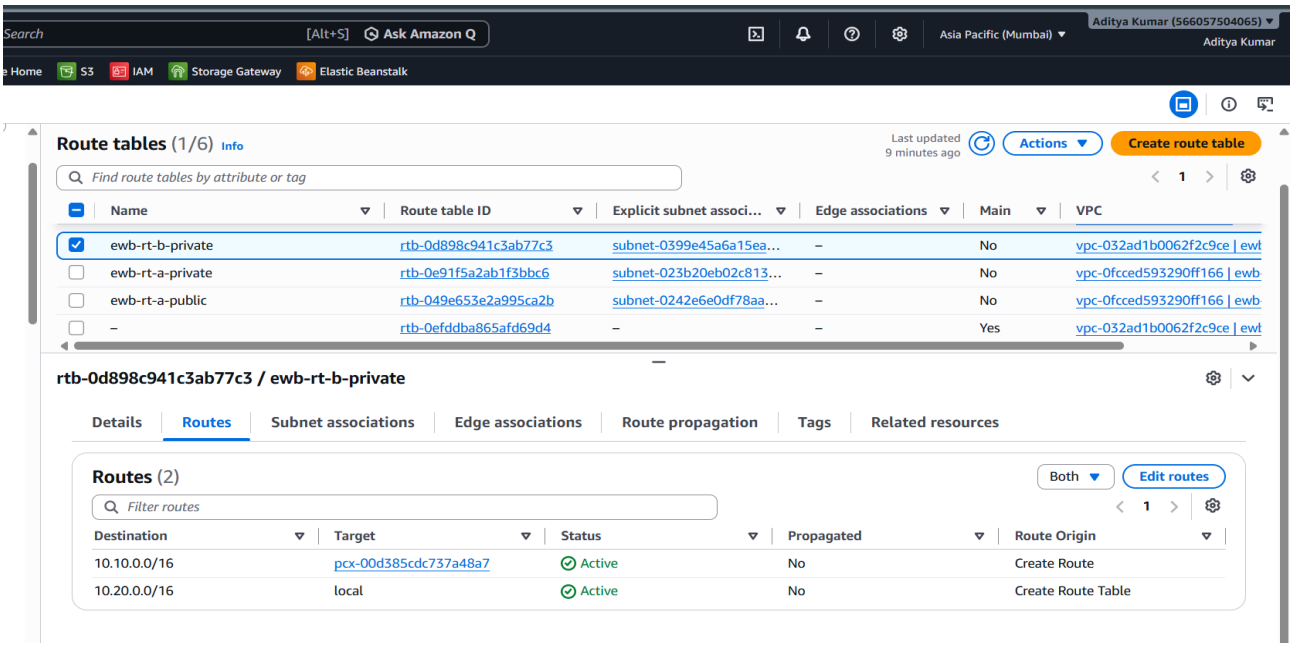
Final VPC A private route table (Add the peering connection route in private)

22-rt-a-private-final.png



Final VPC B private route table

23-rt-b-private-final.png



Successful ping from App A to App B

24-ping-forward (AppA-to-AppB).png

```
--- 10.20.1.108 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3123ms

[ec2-user@ip-10-10-2-105 ~]$ client_loop: send disconnect: Connection reset
client_loop: send disconnect: Unknown error
PS C:\Users\Aditya Kumar> ssh ewb-app-a
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Oct 23 16:32:56 2026 from 10.10.1.108
[ec2-user@ip-10-10-2-105 ~]$ ping -c 4 10.10.1.108
PING 10.10.1.108 (10.10.1.108) 56(84) bytes of data.

--- 10.10.1.108 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3105ms

[ec2-user@ip-10-10-2-105 ~]$ ping -c 4 10.20.1.108
PING 10.20.1.108 (10.20.1.108) 56(84) bytes of data.
64 bytes from 10.20.1.108: icmp_seq=1 ttl=127 time=0.597 ms
64 bytes from 10.20.1.108: icmp_seq=2 ttl=127 time=0.163 ms
64 bytes from 10.20.1.108: icmp_seq=3 ttl=127 time=0.248 ms
64 bytes from 10.20.1.108: icmp_seq=4 ttl=127 time=0.183 ms

--- 10.20.1.108 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3106ms
rtt min/avg/max/mdev = 0.163/0.297/0.597/0.175 ms
```

Successful ping from App B to App A (peering between B to A privately)

25-ping-reverse(AppB-to-AppA).png

```
Connection to 10.10.2.105 closed.
PS C:\Users\Aditya Kumar> ssh ewb-app-b
The authenticity of host '10.20.1.108 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:exJxVVfrssgkPyEyzKC7FbF3WWZGvM4QYcjYP+j2bTQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.108' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-108 ~]$ ping -c 4 10.10.0.0/16
ping: 10.10.0.0/16: Name or service not known
[ec2-user@ip-10-20-1-108 ~]$ ping -c 4 10.10.2.105
PING 10.10.2.105 (10.10.2.105) 56(84) bytes of data.
64 bytes from 10.10.2.105: icmp_seq=1 ttl=127 time=0.168 ms
64 bytes from 10.10.2.105: icmp_seq=2 ttl=127 time=0.194 ms
64 bytes from 10.10.2.105: icmp_seq=3 ttl=127 time=0.169 ms
64 bytes from 10.10.2.105: icmp_seq=4 ttl=127 time=0.186 ms

--- 10.10.2.105 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3114ms
rtt min/avg/max/mdev = 0.168/0.179/0.194/0.011 ms
[ec2-user@ip-10-20-1-108 ~]$
```

SSH access to private App B through bastion (Hostname, Hostname -I and ping to VPC A-private)

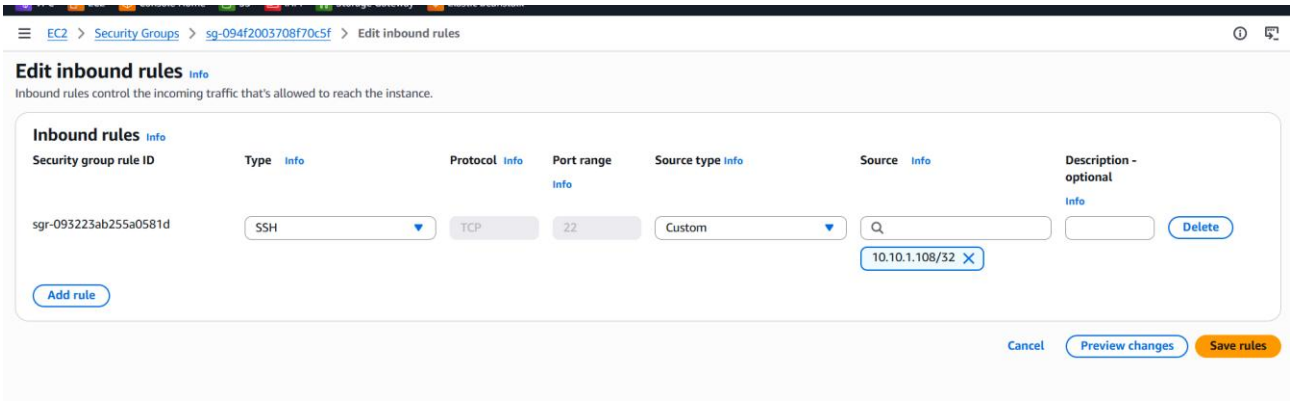
26-ssh-app-b.png

```
PS C:\Users\Aditya Kumar> ssh ewb-app-b
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Oct 23 17:15:50 2026 from 10.10.1.108
[ec2-user@ip-10-20-1-108 ~]$ hostname
ip-10-20-1-108.ap-south-1.compute.internal
[ec2-user@ip-10-20-1-108 ~]$ hostname -I
10.20.1.108
[ec2-user@ip-10-20-1-108 ~]$ ping -c 4 10.10.2.105
PING 10.10.2.105 (10.10.2.105) 56(84) bytes of data.
64 bytes from 10.10.2.105: icmp_seq=1 ttl=127 time=0.167 ms
64 bytes from 10.10.2.105: icmp_seq=2 ttl=127 time=0.176 ms
64 bytes from 10.10.2.105: icmp_seq=3 ttl=127 time=0.191 ms
64 bytes from 10.10.2.105: icmp_seq=4 ttl=127 time=0.185 ms

--- 10.10.2.105 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3156ms
rtt min/avg/max/mdev = 0.167/0.179/0.191/0.009 ms
[ec2-user@ip-10-20-1-108 ~]$
```

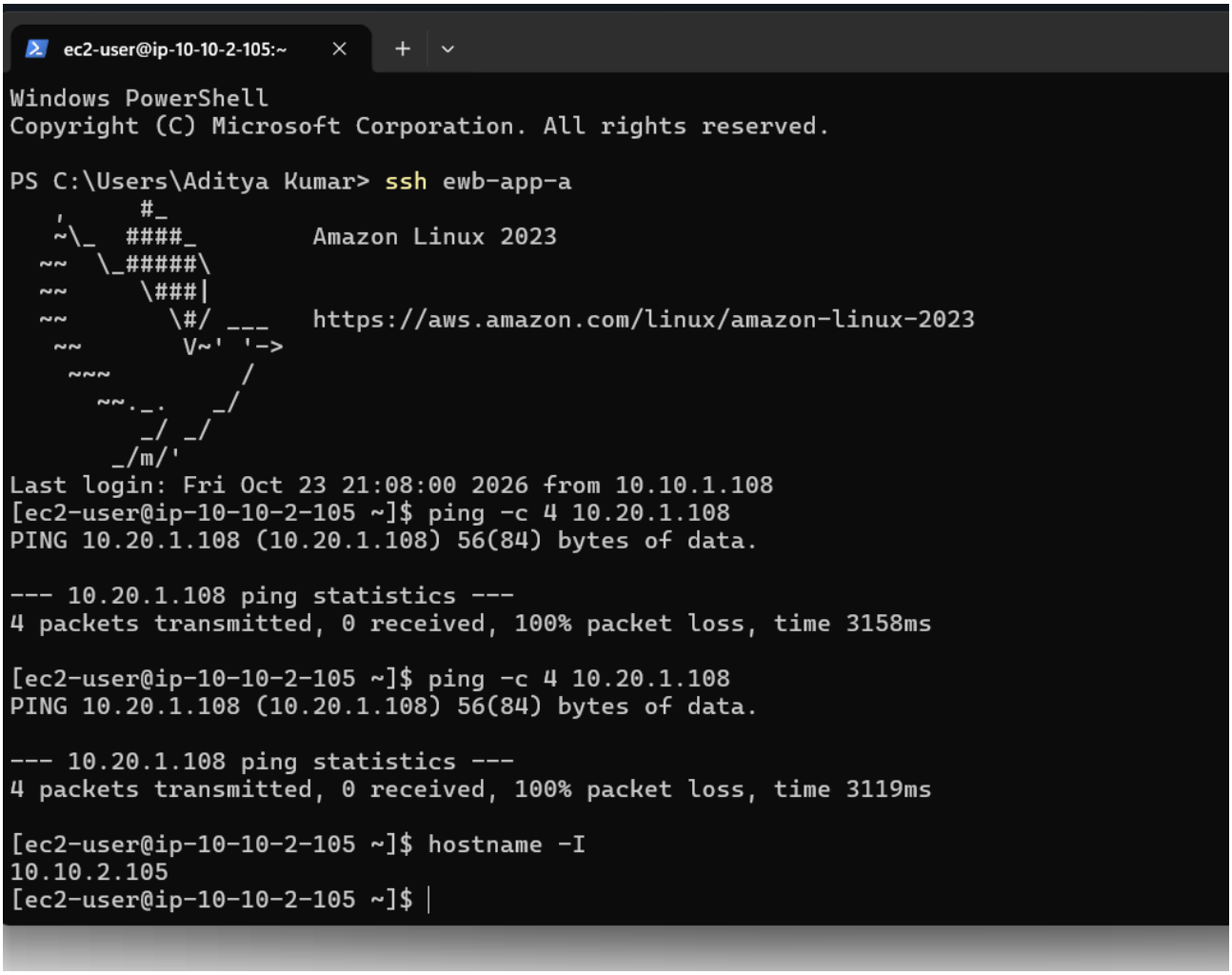
App B ICMP inbound rule removed for controlled failure

27-failure-rule-removed(ICMP from SG).png



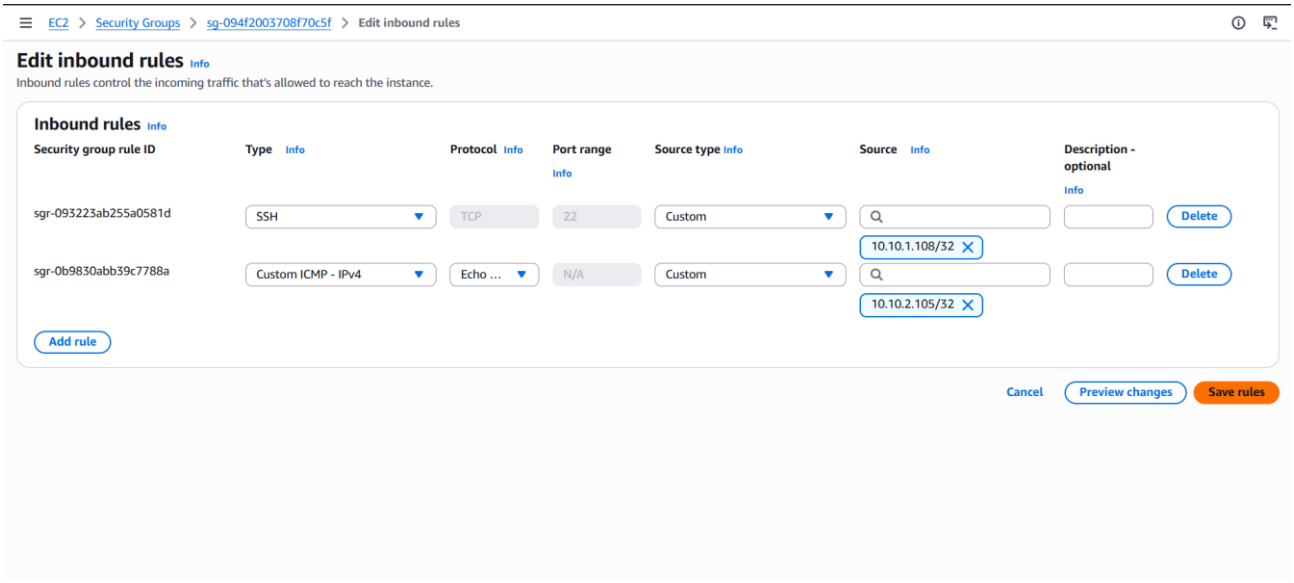
Ping failed during controlled failure

28-failure-ping-fails (From A to B).png



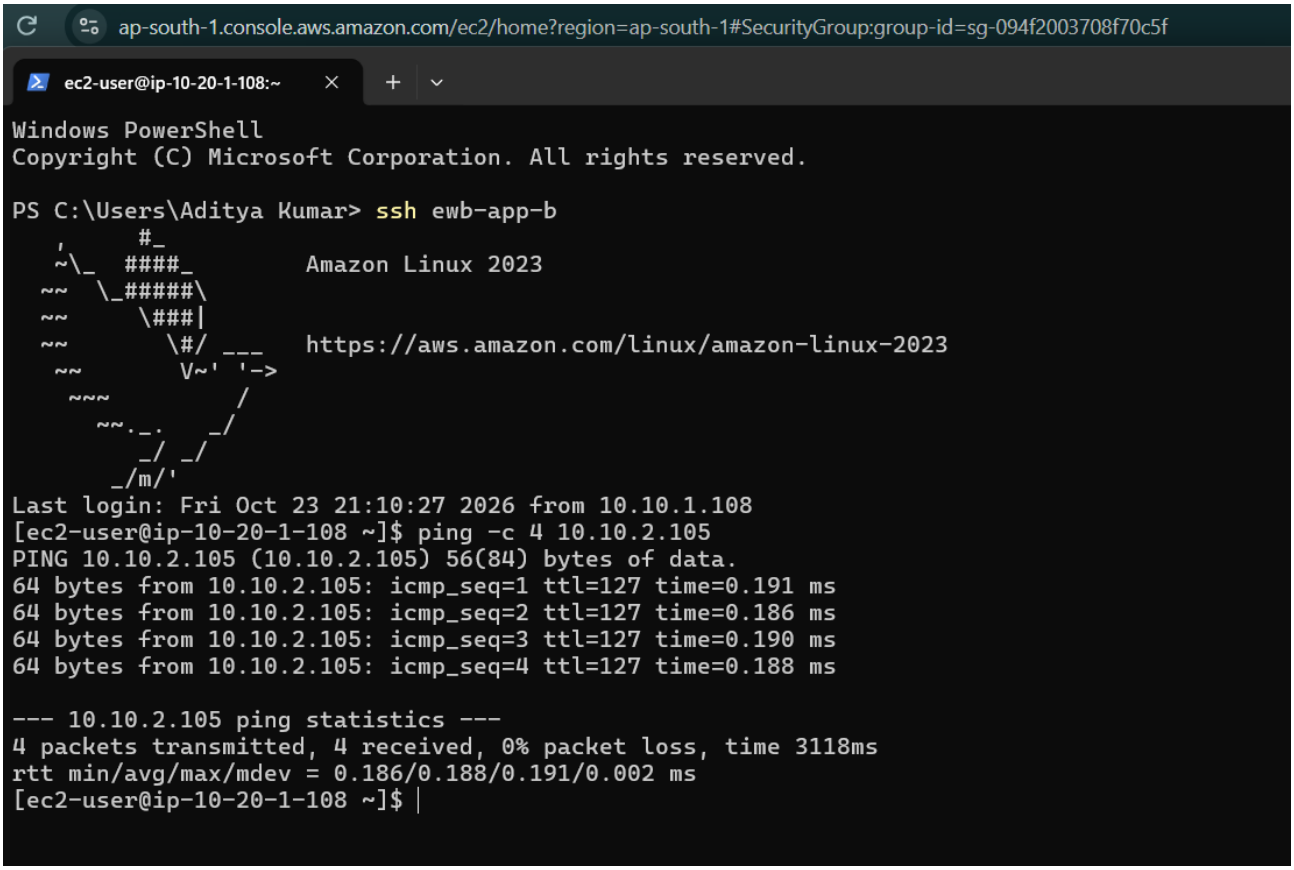
App B ICMP inbound rule restored

29-rule-restored.png



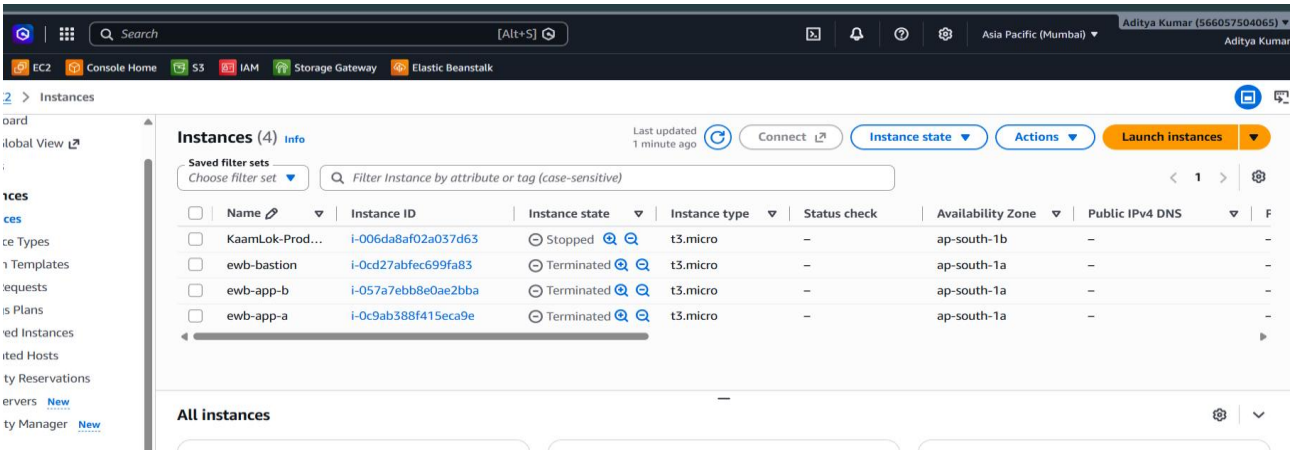
Ping recovered after restoring ICMP rule

30-recovery-ping-ok.png



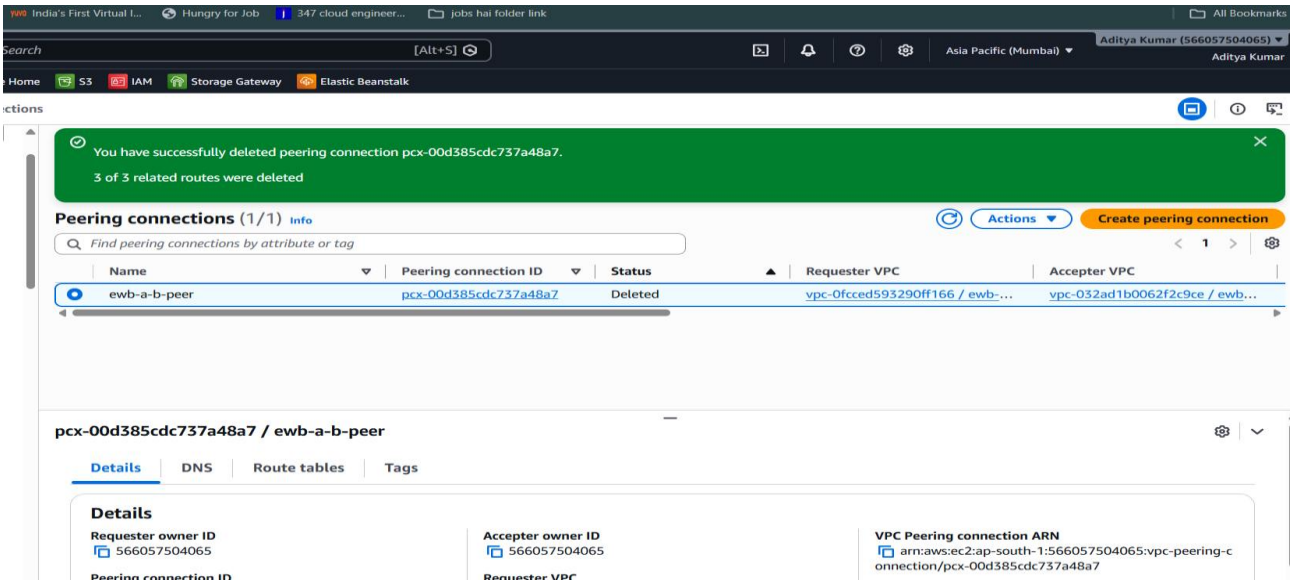
Lab instances terminated (Now we Should to deleted all the resources so no credits will deduct from your aws account and after this all ,once again check the Billing sections in AWS console)

31-cleanup-instances-terminated.png



Peering connection deleted

32-cleanup-peering-deleted.png



Lab VPCs deleted

33-cleanup-vpcs-deleted.png

